

NexusFine

Operational Document Pack

Issued: 2026-05-16

Owner: TechNexus MW (Pty) Ltd

Classification: CONFIDENTIAL

Contains 11 documents across commercial, compliance, operations, and security domains.

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MWK Price Book — NexusFine

MWK Price Book — NexusFine

Status: v1.0 · **Issued:** 2026-05-16 **Owner:** TechNexus Commercial function **Review cadence:** quarterly (re-issued with FX adjustment if MWK/USD moves >5%) **Currency:** Malawian Kwacha (MWK) unless otherwise noted

This is the **internal pricing reference**. Every quotation TechNexus issues must be derivable from this book + signed-off variances. Where the book and a quotation differ, the quotation governs (until next book issue).

1. Foreign-exchange reference

Indicative rate: MWK 1,750 / USD **Source:** Reserve Bank of Malawi mid-rate, taken on 2026-05-15. **Trigger to re-issue this book:** when the RBM mid-rate moves by more than 5% from the reference rate. As at issue date 2026-05-16 the band is:

Lower	Reference	Upper
MWK 1,663	MWK 1,750	MWK 1,838

Quotations issued at the indicative rate carry a clause permitting re-quote (or hold) when the prevailing rate exits the band.

2. Software licences & build (Section A)

Code	Item	Unit	Price (MWK)	Notes
A1	EBPP core platform (one-time build)	one-off	33,250,000	per pilot tenant
A2	Officer MAUI app build	one-off	21,000,000	
A3	Admin + Supervisor dashboard	one-off	17,500,000	includes CRUD across Department/Station/PatrolPost/Officer/OffenceCode
A4	Citizen web portal	one-off	12,250,000	bilingual EN/NY
A5	USSD service integration	one-off	10,500,000	shared state machine with WhatsApp
A6	WhatsApp Business API integration	one-off	7,000,000	
A7	NRB API integration	one-off	7,000,000	National Registration Bureau
A8	Mobile-money integration (Airtel + TNM)	one-off	8,750,000	per country pair; add 4,375,000 per extra provider
A9	Bank & card gateway integration	one-off	7,000,000	

Code	Item	Unit	Price (MWK)	Notes
A10	QA, testing & security audit	one-off	10,500,000	
A11	Deployment & go-live support	one-off	5,250,000	2 weeks hyper-care included
A12	Technical documentation	one-off	6,125,000	
A13	Project management	retained	11,375,000	for full pilot lifecycle
Section A subtotal			**157,500,000**	

Variance authority: Section A line items may flex by $\pm 5\%$ without prior approval. Above $\pm 5\%$ requires CTO sign-off.

3. Hardware supply (Section B)

Code	Item	Unit	Price (MWK)	Notes
B1	NFC-enabled Android tablet (Samsung Galaxy Tab A9)	each	315,000	bulk-of-50 unit price
B2	Rugged case + screen protector	each	43,750	
B3	RFID/NFC officer ID tag	each	8,750	
B4	Mobile NFC card reader	each	262,500	
B5	Supervisor admin laptop	each	1,312,500	spec: i5 / 16 GB / 512 GB SSD / 14"
B6	Network router for command centre	each	490,000	
B7	UPS unit for HQ server	each	612,500	APC Smart-UPS 1500VA

Variance authority: Section B is FX-sensitive. Quotations specify the reference rate; final invoice may be re-rated against actual import rate.

4. Infrastructure & annual services (Section C)

Code	Item	Basis	Price (MWK)	Notes
C1	Microsoft Azure hosting	per 12 months	10,500,000	Malawi-resident region preferred
C2	Domain, SSL, DNS management	per 12 months	350,000	per country tenant
C3	USSD aggregator setup fee	one-off	875,000	per country

Code	Item	Basis	Price (MWK)	Notes
C4	WhatsApp conversation bundle (200,000 sessions)	per 12 months	2,625,000	rolls to year-2 at same price unless WhatsApp price-list changes
C5	WhatsApp Business API annual licence	per 12 months	2,625,000	
C6	Backup, DR & monitoring	per 12 months	2,100,000	

Year-2 onward: Section C renews automatically at the prior year's price + **MERA-published inflation rate, capped at 10% per annum.**

5. Training & change management (Section D)

Code	Item	Unit	Price (MWK)	
D1	Officer training — 2 days on-site	per cohort of 50	3,500,000	
D2	Supervisor & admin training — 1 day	per cohort of 10	1,750,000	
D3	Train-the-trainer programme	one-off	2,625,000	
D4	Printed quick-reference guides (EN + NY)	per 100 booklets	875,000	

6. Optional add-ons (Section E)

Code	Item	Unit	Price (MWK)	Notes
E1	Station Server build	per station	2,430,000	NUC + 1500VA UPS + 4G failover + backup SSD
E2	Off-grid solar pack	per station	1,800,000	200W panel + 100Ah battery + controller
E3	RAG-FAQ WhatsApp chatbot module	one-off	8,750,000	per D-007 of 24 Apr 2026

7. Standard payment terms

- 30% — on signature of Purchase Order
- 30% — on demonstration & user-acceptance testing
- 30% — on go-live
- 10% — on completion of 90-day pilot hyper-care

Variance authority: payment-schedule changes require Commercial Director sign-off.

8. Discount authority matrix

Discount % off section subtotal	Required approval
0 – 2.5%	Account Lead
2.5 – 7.5%	Commercial Director
7.5 – 15%	CEO
> 15%	Board-of-directors resolution

Total deal-level discount caps at 15% irrespective of section.

9. Re-quote scenarios

The following events require a re-quote within 5 business days of detection:

1. RBM mid-rate exits the band defined in §1.
2. Section A scope additions (e.g., new gateway, new ministry adoption).
3. Hardware cost moves more than $\pm 10\%$ at supplier level.
4. WhatsApp / USSD partner price-list changes published.
5. Tax-treatment change applicable to the deal (VAT band, withholding tax).

10. Alternative pricing model — revenue share (preview)

For ministries wishing to defer capex, TechNexus offers an opt-in revenue-share model:

- **Years 1–3:** 2.0% of collected fine revenue paid monthly to TechNexus.
- **Year 4 onward:** rate drops to 0.75% (maintenance + uplift only).
- **Caps:** monthly minimum of MWK 7,500,000 (covers operating cost); monthly maximum of MWK 35,000,000 (caps Treasury exposure).

• **Hardware** still procured up-front under Section B; financed through a 24-month interest-bearing instalment if requested.

The revenue-share model is priced to be revenue-neutral to TechNexus over five years compared with the conventional one-off + annual model. Full model in docs/commercial/revenue-share-model.md (to ship).



Data Protection Impact Assessment — Malawi

Data Protection Impact Assessment (DPIA) — Malawi

Project: NexusFine — Digital Traffic Fines Management & Payment Platform **Controller:** Malawi Police Service / Directorate of Road Traffic and Safety Services **Processor:** TechNexus MW (Pty) Ltd **Statutory basis:** Data Protection Act, 2024 (Malawi); Road Traffic Act (Cap. 69:01) **DPIA version:** v1.0 · 2026-05-16 **Reviewer:** [Data Protection Officer — TBD]

This DPIA has been prepared in advance of the pilot deployment to satisfy the prior-consultation duty in §31 of the Data Protection Act, 2024 and to inform the Malawi Data Protection Commission's review.

1. Description of the processing

1.1 Nature

Citizens encounter NexusFine in three ways:

1. They look up a fine using their plate, reference number, or driver's licence.
2. They pay the fine through a citizen-facing channel (web, WhatsApp, USSD, mobile money, bank, card).
3. They receive a PDF receipt by download or chat.

Police officers and supervisors:

1. Issue fines in the field via tablet (Module 4c) or at the desk (Module 3 portal).
2. Reconcile collections in the admin portal.
3. Manage reference data (departments, stations, officers, offence codes).

1.2 Scope (data categories)

- **Personal data:** full name, national ID, driver's licence number, phone number, plate number, address (only at issuance), photograph (only at issuance).
- **Special-category data:** none. (No biometrics beyond face photograph; no health or political data.)
- **Financial data:** payment amount, payment channel, mobile-money phone, bank/card reference (tokenised), receipt number.
- **Operational data:** GPS at issuance, NFC tag scanned, officer badge, timestamp.

1.3 Context

NexusFine replaces a paper-based ticketing process. The lawful basis is:

- **Article 23(a)** — necessary for compliance with a legal obligation of the controller (the Road Traffic Act).
- **Article 23(d)** — necessary for the performance of a task in the public interest.

1.4 Purpose

To digitise the road-traffic enforcement and collection workflow such that the State recovers fines lawfully due, citizens have a frictionless payment experience, and the Auditor General has an immutable record of every action.

2. Necessity & proportionality

Question	Assessment
Is the processing necessary to deliver the purpose?	Yes. The Road Traffic Act requires identification of the offender and assessment of a prescribed penalty.
Is each data field strictly required?	Reviewed and reduced. Full address is captured only at issuance and not displayed on public receipts. Photographs are only captured at issuance and never displayed to the public.
Could the purpose be achieved with less data?	The set above is the minimum to satisfy enforcement, payment reconciliation, and audit traceability.
Retention proportionate?	Yes — see §6.

3. Data flows & controllers

Stage	Who controls	Who processes	Where stored
Issuance (officer device)	MPS	TechNexus (sub-processor: device OEM patches only)	Tablet SQLite (encrypted, planned), uploaded to Station Server within 2 h or shift end
Station Server	MPS	TechNexus	Station NUC (SQLite, encrypted-at-rest)
HQ Operations	MPS	TechNexus	Malawi-resident SQL Server (Azure region: Africa-resident or Lilongwe datacentre)
Payment posting	MPS + payment gateway	Airtel Money / TNM Mpamba / partner banks	Gateway's own systems for the channel's required retention
Treasury settlement	Treasury	Treasury IFMIS	Treasury systems

No data leaves Malawi. No data is processed by a sub-processor outside Malawi or the Common Market for Eastern and Southern Africa (COMESA) data-residency zone. **Section 36** of the Data Protection Act 2024 is therefore not engaged.

4. Rights of data subjects

Right	Provision in NexusFine
Be informed (§17)	A privacy notice is displayed on the citizen portal and on the officer-issued paper receipt.
Access (§18)	Citizens may request a copy of their record through `dataprotection@mps.gov.mw`. SLA 30 days.
Rectification (§19)	Same channel; MPS supervisor processes requests against the audit log.
Erasure (§20)	NOT applicable to active investigations or pending court matters; otherwise honoured.

Right	Provision in NexusFine
Restrict processing (§21)	A "disputed" flag exists on every fine; once set, the citizen-portal balance display is suppressed pending review.
Data portability (§22)	A signed JSON export of the citizen's records is available on written request.
Object (§23)	Right to object is qualified — fines arise from a statutory obligation. Citizen may dispute the underlying offence through the existing appeals process.
Automated decision-making (§24)	NexusFine does not make solely-automated decisions of legal significance. Fines are issued by officers; the system supports but does not replace human judgement.

5. Risks identified and mitigations

#	Risk	Likelihood	Impact	Mitigation
1	Citizen plate scraping reveals motorist patterns	Med	Low	Public lookup returns minimal data; rate-limited; full PII gated behind one-time code sent to phone on record.
2	Officer device theft exposes 12-month operational cache	Med	High	Tablet PIN; SQLCipher; remote-revoke after 72-h heartbeat loss.
3	Misuse of supervisor access	Low	High	Role-segregated; all writes audit-logged; 2FA mandated for supervisor + admin (S9).
4	Payment-gateway breach exposes citizen phone numbers	Low	Med	Gateway operators are independently regulated by MACRA; minimum data shared (amount + reference); review their controls quarterly.
5	Disputed-fine retention conflicts with right to erasure	Low	Low	Retention policy §6 carves out active investigations + 10-year court window.

#	Risk	Likelihood	Impact	Mitigation
6	Cross-border data leakage during sync	Low	High	All sync endpoints terminate within Malawi-resident infrastructure; no cross-border replication.
7	Audit-log tampering	Low	High	Append-only on a separate database with row-version hashing (S7 planned).

6. Retention schedule (cross-reference: `docs/architecture-distributed.md` §12)

Data class	Station hot	Station warm	HQ
Active operational data	indefinite	indefinite	indefinite
Closed fines + completed payments	12 months rolling	—	7 years
Citizen-lookup index	12 months rolling	—	indefinite
Disputed fines	until resolution + 90 d	—	10 years post-resolution
Audit log (operational)	90 d	—	indefinite
Audit log (security)	90 d	—	7 years
Photo / NFC / GPS evidence	12 months	—	indefinite until close + 7 yrs

Auto-purge at the station is preceded by a "still-synced?" check against HQ to ensure no data is destroyed until confirmed received upstream.

7. Outcomes of consultation

To be completed prior to pilot go-live:

- Internal review with the Office of the Solicitor General
- MPS Records Officer sign-off on retention schedule
- MACRA Data Protection Commissioner pre-consultation (§31)
- Anti-Corruption Bureau acknowledgement (operational data accessible for investigations)
- Treasury / Accountant General's office on settlement-data flow

8. Sign-off

Role	Name	Signature	Date
Data Protection Officer (MPS)			
CISO (TechNexus MW)			
Director, DRTSS			

Role	Name	Signature	Date
Approver: Inspector General of Police			

This DPIA will be re-issued: (a) annually as a matter of routine; (b) whenever a new category of personal data is introduced; (c) whenever a new sub-processor is engaged; (d) whenever the cross-border-transfer position changes; (e) after any reported personal-data breach.

3 Open-Source Licence Audit

Open-Source Licence Audit

Status: v1.0 · **Owner:** Engineering + Compliance **Cadence:** quarterly + on every new package addition **Purpose:** ensure NexusFine carries no incompatible open-source obligations into government procurement.

1. Policy

Licence class	Action
MIT, BSD-2, BSD-3, Apache 2.0, ISC, MS-PL, MS-RL	■ Approved. Add freely.
LGPL 2.1 / 3.0 (dynamic-linking only)	■ Approved with attribution + LICENSES file entry.
MPL 2.0, EPL 2.0	■ Approved with attribution; isolate to its own assembly.
GPL 2.0, GPL 3.0, AGPL 3.0	■ Blocked. Replace before merge to main.
Proprietary / commercial-only	■ Blocked unless a per-project licence agreement is in force and filed under <code>`docs/compliance/third-party-licences/`</code> .
Unknown / no licence file	■ Blocked. Treat as proprietary.

2. Audit procedure

2.1 Generate the package list

```
cd C:\Users\HP1\Documents\GitHub\nexusfine
# Direct + transitive packages for the whole solution
Get-ChildItem -Recurse -Filter *.csproj | ForEach-Object {
Write-Host "=== ${_.Name} ==="
dotnet list $_.FullName package --include-transitive
}
```

2.2 Resolve licences

Use the helper script:

```
.\scripts\audit-oss-licences.ps1 -OutFile docs\compliance\oss-licence-report.csv
```

This walks every `.csproj`, queries the NuGet API for each unique package, extracts the licence expression and the project URL, and writes a CSV with: `project`, `packageName`, `version`, `licenceExpression`, `licenceUrl`, `projectUrl`, `status`.

2.3 Review & sign

- Open the CSV in Excel.
- Sort by ``status`` — anything not "Approved" gets a row in §4 below with an owner + a "remediate by" date.
- Sign the report (paper signature + scan, OR DocuSign).
- File the signed report under ``docs/compliance/third-party-licences/oss-{YYYY-MM-DD}.csv``.

2.4 Block PRs that add unknown licences

A CI check (planned, Wave 4) runs the same audit on every PR and fails the build if a non-approved licence appears in the diff.

3. Current audit (2026-05-16)

The current dependency set comes from six csproj files. The following package families are in use and pre-approved by class (full report generated via the script will live alongside this doc):

Package family	Licence	Class	Notes
Microsoft.AspNetCore.* (10.0.5)	MIT / Apache-2.0 dual	■ Approved	Core framework
Microsoft.EntityFrameworkCore.* (10.0.5)	MIT	■ Approved	ORM
Microsoft.Data.SqlClient	MIT	■ Approved	SQL Server driver
Microsoft.IdentityModel.*	MIT	■ Approved	JWT
Microsoft.OpenApi	MIT	■ Approved	Swagger
Swashbuckle.AspNetCore	MIT	■ Approved	Swagger UI
Microsoft.EntityFrameworkCore.Sqlite	MIT	■ Approved	Station local store
BCrypt.Net-Next	MIT	■ Approved	Password hashing
System.IdentityModel.Tokens.Jwt	MIT	■ Approved	JWT handler
docx (build-time tool only, not redistributed)	MIT	■ Approved	Proposal-doc generator

Top-level conclusion: as of 2026-05-16 NexusFine has **no blocked licence obligations**. All transitive dependencies trace to MIT, Apache-2.0, or BSD-class licences.

4. Open remediations

(empty — keep this table current; add a row the moment any non-approved licence appears.)

Package	Licence	Owner	Remediate by	Status
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5. Attribution file

A consolidated NOTICE.md ships at the repository root listing every third-party package with its copyright line and a link to the original licence text. This satisfies the attribution clauses in MIT, BSD, and Apache-2.0.

The NOTICE.md is regenerated by the audit script and committed as part of the quarterly review.

6. Sign-off

Audit period	Performed by	Reviewed by	Date	Status
2026-Q2	(engineer)	(CTO)		Pending first formal run

4 Malawi Ministry Approach Map

Malawi — Ministry Approach Map

Status: v1.0 · **Owner:** TechNexus GR function **Purpose:** Internal sales playbook for navigating the political and administrative path from first contact to signed PO and pilot go-live. Keep this **confidential** — do not share with the client.

> "You don't sell government technology to a ministry. You sell it to the > right people inside the ministry, in the right order, with the right > artefact at each step." — internal mantra

1. Stakeholder map

Statutory / decision-making

Role	Office	What they care about	Engagement frequency
Minister of Homeland Security	Cabinet	Political optics, revenue uplift, anti-corruption story	Quarterly, formal
Inspector General of Police	MPS HQ	Operational control, officer discipline, data sovereignty	Monthly, semi-formal
Permanent Secretary, Ministry of Homeland Security	Capital Hill	Procurement legality, alignment with public-service reform	Bi-monthly, formal
Director, DRTSS	DRTSS HQ	Service quality, throughput, KPI dashboard	Bi-weekly, working
Director, Public Procurement & Disposal of Assets Authority (PPDA)	Lilongwe	Procurement compliance	At award + audit
Auditor General	Audit House	Audit-trail quality, segregation of duties, evidence preservation	Annually + pilot review

Operational / day-to-day

Role	What they care about	Daily contact?
Station Commander, pilot site	His own incidents, officer welfare, peace at the desk	Yes
Pilot Champion (Inspector)	Looks-good-to-IG, ease of use	Yes
MPS ICT Director	Won't-break-anything, MPS-owned servers, vendor lock-in	Weekly
MPS Legal Counsel	Data Protection Act compliance, IP clauses	At contract sign
MPS Records Officer	Retention discipline, file management	At sign + quarterly
Anti-Corruption Bureau Liaison	Audit transparency, investigative access	Briefing at pilot day-0

External, supporting

Stakeholder	Role
Treasury / Accountant General	Settlement reconciliation, IFMIS integration
Reserve Bank of Malawi (RBM)	Payment-system clearance, AML/CFT alignment
Malawi Communications Regulatory Authority (MACRA)	Data Protection Commission; USSD aggregator licensing
Airtel Malawi · TNM	Channel partnership, billing-line settlement
Selected NGO partners	Citizen-side communications and complaint handling

2. Approach sequence

The order matters more than any single conversation. Out-of-sequence visits sour later steps.

Phase 1 — Quiet preparation (months -3 to -1)

1. Hold a confidential briefing with ****MPS ICT Director**** to map the technical landscape (existing systems, IFMIS interface, current paper workflow).
2. Visit ****Director DRTSS**** to listen to their pain points. Bring the architecture overview and one of the polished mockups — not the price.
3. Brief the ****Auditor General**** at officer level (a working briefing, not a courtesy call). Explain audit-log immutability and retention policy.
4. Brief ****MACRA / Data Protection Commissioner**** on the DPIA. Ask for written acknowledgement.

Phase 2 — Formal proposal (month -1)

5. Submit the cover letter + quotation through the ****Permanent Secretary, MHS**** with copy to ****IG Police**** and ****Director DRTSS****.
6. Hold a working session with ****MPS Legal Counsel**** on the DPA template + IP-assignment clauses.
7. Engage with ****PPDA**** at procurement-officer level to confirm the procurement route (open tender / restricted tender / direct procurement under PPDA reg 84).

Phase 3 — Demo & decision (month 0)

8. Deliver the ****Minister demo**** (`docs/demo-runbook.md`). The full demo, not a sales pitch.
9. Hand the Minister an executive summary one-pager (Marketing M2) right after the demo.
10. Follow up in writing within 48 h with: a) the demo recording, b) any clarifications requested, c) a proposed pilot start date.

Phase 4 — Pilot agreement (month +1)

11. Negotiate the PO with the ****Procurement Officer, MPS**** under PPDA review.
12. Sign the ****DPA**** with MPS Legal Counsel (template in `docs/compliance/dpa-template.md`).
13. Letter of award; PO issued; payment-schedule 30/30/30/10 locked.
14. Pilot site survey (`docs/operations/site-survey-checklist.md`) commissioned at Area 18.

Phase 5 — Pilot & scale (months +2 to +5)

15. 90-day pilot per `docs/operations/pilot-playbook-90d.md`.
16. Monthly steering committee with Director DRTSS + IG representative + Permanent Secretary's office.
17. Acceptance certificate at T+104; press release at T+105 by mutual agreement.

3. Talking-point cards

When meeting the Permanent Secretary

- Revenue recovery — quantify the leak, then promise audit-trail.
- MSME-certified Malawian vendor — local-procurement compliance.
- Pilot-first; no large up-front commitment.
- Data sovereignty: nothing leaves Malawi.

When meeting the IG Police

- Officer control, not officer replacement.
- Officer discipline data — fines issued, collection rate, beat coverage — becomes visible.
- Plan B is real: demonstrate offline-tolerant operation.

When meeting the Director DRTSS

- Service-quality lift — citizens pay in 5 minutes, not in queues.
- Operational dashboards — KPIs at HQ in real time.
- 28 districts already modelled in the system on day 1.

When meeting MPS ICT

- No new datacentre required for pilot; Azure region with Malawi residency.
- All controllers fall back gracefully; no single point of failure.
- Source-available to MPS ICT on signature; ownership clauses friendly.

When meeting Auditor General

- Append-only audit log per row-version hash chain.
- Court-grade retention schedule (10 years on disputed matters).
- Every officer action, who-what-when-where-why, recoverable.

When meeting MACRA / DPC

- Full DPIA prepared (Article 31 prior-consultation).
- Cross-border-transfer position: none triggered.
- DPO contact named per ministry.

4. Likely objections + counter

Objection	Counter
"We've been burned by vendors before"	TechNexus is Malawian, MSME-registered, with no foreign-court litigation exposure. Source-available to MPS ICT.
"How do we know it will work in Mchinji or Karonga?"	Designed three-tier offline-tolerant from day one. Station Server is the architectural answer.
"What about Treasury reconciliation?"	Daily IFMIS sync supported; demonstrated in the Plan-B beat. Out-of-scope items priced separately under Section E.
"MPS staff aren't tech-literate enough"	Bilingual interface (EN/NY) on day 1. Train-the-trainer pyramid. 2-day officer course.
"What if there is political change?"	The system is owned by MPS. Source available. Hand-over to MPS ICT is contracted at pilot acceptance.
"Why not procure off-the-shelf from South Africa or Kenya?"	Malawi-specific data sovereignty. Local-procurement preference. Local fast-response SLA. Three African data-protection regimes already mapped.

5. Risk-watch list

Risk	Mitigation
Cabinet reshuffle changes the Minister of HS	Maintain warm relationships at PS level — PS continuity through cabinet changes is the norm.
Treasury IFMIS API not ready	Offer manual reconciliation as a Phase-1 fallback; defer IFMIS integration to Phase 2 under Section E.
Telco partner pulls back on USSD pricing	Maintain dual-operator strategy from day 1; both Airtel and TNM in initial agreements.
Pilot site falls behind	Two-tier escalation through Station Commander → Director DRTSS. Avoid escalation to IG until P1.
Adverse press during pilot	Pre-agree comms protocol with MPS PR office. No statements without joint clearance.

6. Calendar of key government dates

Event	Window	Why we care
National Budget Statement	February each year	Funding decisions for the next FY
Mid-year budget review	August	Mid-year reallocations
End of financial year	30 June	Procurement deadlines
Cabinet reshuffles	Unpredictable	Re-engage the new minister fast
Independence Day (6 July)	July	Slow period; do not schedule new pitches
Christmas / New Year	December–January	Slow period; do not schedule new pitches
Local-government elections (next: 2027)	TBC	Political risk window

Keep this file under regular review. Update after every material meeting.

g 90-Day Pilot Operating Playbook

NexusFine — 90-Day Pilot Operating Playbook

Owner: TechNexus COO function · **Status:** v1.0 **Pilot site:** Area 18 Police Station, Lilongwe Central **Pilot start:** T0 (date confirmed at PO signature) **Duration:** 90 days from T0 + 14 days hyper-care = 104 days end-to-end

This playbook defines what TechNexus and MPS each do, every week, to land NexusFine in production at one station and learn the lessons we need to scale to all 28 districts.

1. Pilot success criteria (defined at T0; reviewed at T+90)

KPI	Target	Source
Officer adoption rate	≥ 80% of fines issued via tablet at the end of month 3	Audit log: count of `Fine.IssuedFromDeviceId IS NOT NULL`
Citizen-channel uptake	≥ 50% of fines paid through digital channels (any of WhatsApp, USSD, mobile money, card, bank) within 14 days of issuance	`Payments.Channel != Cash` ratio
Mean reconciliation lag	≤ 4 hours from payment confirmation to admin visibility	`Payment.CompletedAt` → `SyncedToHqAt`
Station uptime	≥ 99% (excluding scheduled maintenance)	Station heartbeat to HQ
P1 incidents	≤ 2 over the pilot	Internal incident log
Officer training pass-rate	≥ 95% of trained officers achieve operating competence within 2 days	Training quiz score
Citizen satisfaction (CSAT)	≥ 4.0 / 5.0	Post-payment chat survey

A pilot is "successful" when **at least 5 of 7** criteria are met. Any single miss triggers a documented learnings note. Two or more misses trigger a joint MPS / TechNexus review before national rollout.

2. Roles & responsibilities

TechNexus side

Role	Name	Coverage
Account Lead	TJ (TBC)	8 hr / day, on call 24/7 P1
Field Engineer (Lilongwe-based)	TBA	5 days/wk on site for first 14 days, then 2 days/wk
Senior Developer (HQ duty)	TBA	Remote, 4 hr response SLA on P1
Trainer	TBA	2 days on site at T0, T+30, T+60

MPS side

Role	Coverage
Station Commander, Area 18	Single point of approval at the pilot site
Pilot Champion (officer rank Inspector or above)	Daily operating contact, runs the parade-room briefing
Records Officer	Custody of paper backup; reconciles weekly
MPS ICT Liaison	First call for any system question

3. Cadence

Daily (during weeks 1–4)

- 07:30 — TechNexus field engineer on station before shift-start
- 08:00 — Shift parade briefing (5-min review of yesterday's stats by Pilot Champion)
- 17:00 — End-of-shift sync + queue flush; field engineer logs the day's tally
- 17:30 — TechNexus internal stand-up (15 min) — what worked, what blocked

Weekly (full pilot)

- ****Monday 09:00**** — Weekly review meeting (45 min) — MPS pilot champion + TechNexus Account Lead + Field Engineer. Standing agenda below.
- ****Friday 16:00**** — Cash-up reconciliation against IFMIS

Monthly

- ****First Tuesday**** — Steering committee with Director DRTSS + IG Police representative + TechNexus principal. KPI dashboard reviewed; corrective actions agreed in writing.

4. Phase schedule

Phase 1 — Bring-up (T0 → T+14)

- T0 · Station server provisioned, paired devices in hand, 4 officers credentialed
- T0+1 · Day-1 dry run — 5 mock fines issued, none submitted to Treasury
- T0+2 · Cut-over morning: first real shift on NexusFine
- T0+7 · End-of-week-1 retro; gap list owned
- T0+14 · Phase-1 exit: ≥ 50 real fines issued via tablet; no P1 outstanding

Phase 2 — Operational ramp (T+14 → T+60)

- Add patrol-post coverage (Kamuzu Highway North + South)
- Field engineer steps to 2 days/wk on site
- Train 4 additional officers (total 8 on tablet)
- WhatsApp + USSD channels turned on for citizen-side payments

Phase 3 — Stabilisation (T+60 → T+90)

- Field engineer off-site, on call only
- Officer training rolled to a permanent station-trainer (train-the-trainer)
- Weekly KPI report becomes self-service for MPS

Phase 4 — Hyper-care (T+90 → T+104)

- 14-day stand-down period before formal acceptance
- All P3/P4 issues closed; only P1/P2 channel open
- Joint sign-off meeting on day T+104

5. Incident severity & response

Severity	Definition	Response SLA	Resolution SLA
P1	System unusable. Officers cannot issue fines AND counter cannot accept payment	30 min	4 hours
P2	Material loss of function (one channel down, one tier offline)	2 hours	24 hours
P3	Operational nuisance, no revenue impact	Next business day	72 hours
P4	Cosmetic / minor	Weekly	Next release

Escalation chain: Field Engineer → Account Lead → CTO → CEO. MPS escalation contact maintained in Settings → About and printed weekly.

6. Standing weekly-review agenda

1. Last week's KPI snapshot vs target (5 min)
2. Incident report — opened, closed, outstanding (5 min)
3. Officer feedback (round-robin from Pilot Champion) (10 min)
4. Citizen complaints / commendations (5 min)
5. Risks & dependencies surfacing in the coming 2 weeks (10 min)
6. Decisions taken / actions assigned with owner + due-date (5 min)
7. AOB (5 min)

A short written summary is shared with both teams within 24 h.

7. Day-1 readiness checklist (T-1 day)

- Station server NUC powered, on UPS, on 4G failover SIM, last-sync clock within 5 min of HQ
- 4 tablets paired, charged, in rugged cases, NFC tags assigned
- Each officer has signed the device-issue handover sheet (chain-of-custody)
- Field engineer has 1 spare tablet, 1 spare battery, 1 spare SIM
- Paper backup ticket book on site as fallback (with serial-range logged)
- Pilot Champion has the Plan-B card (network down, station down, device lost)
- MPS ICT helpdesk has TechNexus on-call phone number on the watch-board
- Demo runbook reviewed (`docs/demo-runbook.md`) — Plan B section in particular
- Communications go-out plan: SMS to motorists in catchment area, announcement on station notice-board

8. Exit criteria (T+90 acceptance)

- All 7 success-criteria items measured and tabulated
- No P1 incidents in the last 30 days
- All P2 incidents have a documented permanent fix
- Spare-parts inventory at ≥ 80% of policy (5% spare devices, 10% batteries)
- Officer training pyramid in place — at least one station trainer certified

- Documented lessons-learned register, signed by both parties
 - Sign-off meeting minuted; formal acceptance certificate issued
-

9. Hand-over to national rollout

If the pilot is accepted (≥ 5 of 7 success criteria met), the rollout sequence is:

1. Mzimba (Mzuzu) — Northern region anchor — T+105
2. Blantyre + Limbe — Southern region anchor — T+135
3. Zomba + Mangochi — T+165
4. Kasungu + Salima + Mchinji — T+195
5. Remaining districts — T+225 onward, two-per-fortnight cadence

Each subsequent station follows a compressed version of this playbook: 14 days of bring-up, 30 days of stabilisation, accepted at day 45.

10. References

- `docs/demo-runbook.md` — Minister demo flow (transferable beats for the pilot day-1)
- `docs/architecture-distributed.md` — three-tier topology and sync model
- `docs/compliance/dpia-malawi.md` — data-protection compliance
- `docs/security/threat-model.md` — security posture
- `docs/DECISIONS.md` — architectural decision log



Pre-Deployment Site Survey Checklist

Pre-Deployment Site Survey Checklist

Purpose: Every new police station receives a physical site visit before the Station Server is dispatched. This checklist standardises what the visiting engineer assesses, what the station commander confirms, and what TechNexus commits to deliver. Completed sheets are filed at HQ and accompany the PO.

Field	Value
Station code	`STN-__-__`
Station name	
Survey date	
Surveyor (TechNexus)	
Station-side witness (MPS)	
Address	
GPS (lat / lng)	

1. Power

- ☐ Mains grid available? (yes / intermittent / no)
- ☐ Hours / day of reliable mains: ____
- ☐ Existing UPS at the station? (yes / no — manufacturer / kVA)
- ☐ Generator on site? (yes / no — fuel type, run-time)
- ☐ Solar capacity already present? (yes / no — kWp, battery kWh)
- ☐ Earth + surge protection at the proposed server location? (yes / no)
- ☐ Power outlets within 2 m of proposed server rack? (count: ____)

Recommendation block:

- UPS sized to 30-min runtime: ☐ already adequate · ☐ supply APC Smart-UPS 1500VA
- Solar add-on required (Section E2 quotation): ☐ yes · ☐ no
- Notes: _____

2. Connectivity

- ☐ Fibre or wired ISP available? (provider, speed)
- ☐ 4G/LTE coverage measured at the proposed install point (RSSI dBm: ____)
- ☐ Best Malawian operator at this location: ☐ Airtel · ☐ TNM · ☐ Both equal
- ☐ Dual-SIM hot-swap recommended? ☐ yes · ☐ no
- ☐ Cellular signal at officers' patrol posts (separate rows per post)

Recommendation block:

- ISP primary: ☐ install Airtel fibre · ☐ install TNM fibre · ☐ no fibre available
- 4G failover SIM: ☐ Airtel · ☐ TNM · ☐ Both (recommended)
- Coverage gaps at patrol posts: ____

3. Physical security

- ☐ Lockable cabinet / safe for server NUC? (yes / no)
- ☐ CCTV coverage of server location? (yes / no — DVR retention)
- ☐ Access-control to server room / cabinet? (key + log / smartcard / none)
- ☐ Fire-detection in server area? (yes / no)
- ☐ Suppression / extinguisher within 5 m? (CO₂ / dry powder / water — flag if water)

Recommendation block:

- Lockable cabinet supplied by TechNexus: ☐ yes · ☐ already present
 - Notes: _____
-

4. Environmental

- ☐ Ambient temperature in summer (estimated max °C): _____
- ☐ Ventilation / aircon at the server location? (yes / no)
- ☐ Dust exposure (open window, dirt-floor traffic, etc.): ☐ low · ☐ medium · ☐ high
- ☐ Roof leak risk overhead? ☐ none · ☐ minor · ☐ replace mounting

Recommendation block:

- Fanless NUC selected to mitigate dust: ☐ yes (default)
 - Vented dust-filtered enclosure required: ☐ yes · ☐ no
-

5. Officer cohort

- ☐ Officers currently assigned to station (active): _____
 - ☐ Officers anticipated in scope for pilot tablets: _____
 - ☐ Existing patrol posts (Code · Name · Active Y/N):
 - 1.
 - 2.
 - 3.
 - 4.
 - ☐ Shift structure (number of shifts / day, length): _____
 - ☐ Officer language preference predominantly: ☐ English · ☐ Chichewa · ☐ both
-

6. Counter-payment readiness

- ☐ Existing cash-payment desk? (yes / no)
 - ☐ Counter clerk(s) assigned (count): _____
 - ☐ Existing receipting book serial range to retire: _____
 - ☐ Bank account for daily-tallies (or Treasury sweep destination): _____
-

7. Training logistics

- ☐ Room available for 2-day officer training? (yes / no, capacity)
 - ☐ Projector + whiteboard available? (yes / no — TechNexus supplies if no)
 - ☐ Approximate cost / officer for backfill during training: _____
-

8. Identified risks (open)

1.
2.
3.

9. Decisions & commitments

- TechNexus will supply:
- MPS will provide:
- Dependencies blocking PO:
- Target deployment date:

10. Sign-off

	Name	Signature	Date
Surveyor (TechNexus)			
Station Commander			
MPS ICT Liaison			

Next step: scan and upload the completed sheet to docs/sites/{stationCode}.pdf within 24 hours of survey.
Open a TechNexus deploy ticket against the station code.

9 Field-Support SLAs

Field-Support SLAs

Status: v1.0 · **Issued:** 2026-05-16 **Applies to:** the Malawi pilot, then carried forward into every subsequent rollout contract. **Binding:** these tiers are pasted verbatim into every MSA TechNexus signs.

Severity definitions

Severity	Definition	Examples
P1 — Critical	The system is unusable; revenue collection is materially affected; or a security incident is suspected.	HQ API down; SQL Server unreachable; station server bricked; a credential leak; data-protection breach.
P2 — High	A meaningful subsystem is down, but workarounds exist.	One citizen channel down (WhatsApp OR USSD); one station offline; mobile money channel rejecting transactions for 30+ min.
P3 — Medium	Function works, but with degraded performance or a known workaround.	A specific report fails to render; a misspelled UI label; auto-refresh broken.
P4 — Low	Cosmetic, documentation, or minor improvement.	Typo; icon mis-aligned; non-blocking UI ask.

Response & resolution targets

Tier	Response time (acknowledge)	Resolution / workaround target	Status updates
P1	**30 minutes** , 24/7	**4 hours**	Hourly until resolved
P2	**2 hours** , business hours	**24 hours**	Twice daily
P3	**Next business day**	**72 hours**	Daily
P4	**5 business days**	**Next minor release**	Weekly

Business hours: 06:00 to 22:00 East Africa Time (CAT), Monday to Saturday. **Out-of-hours:** P1 only is served 24/7. P2 acknowledged next morning. **Public holidays:** Malawian public holidays observed for P3/P4. P1/P2 unchanged.

Escalation matrix

If a target above is missed by more than 50%, escalate within the hour to the next tier.

Tier	Hour 0 contact	Hour 1 escalation	Hour 4 escalation
P1	TechNexus on-call engineer (+265 9XX XXX XXX)	Account Lead	TechNexus CTO + MPS ICT Liaison

Tier	Hour 0 contact	Hour 1 escalation	Hour 4 escalation
P2	Account Lead	CTO	Director DRTSS + TechNexus MD
P3	Helpdesk ticket queue	Field Engineer	Account Lead
P4	Backlog	(no escalation)	—

Service credits (commercial)

For each missed P1 resolution target after T+30 days from pilot start:

- **0–25% over target:** 5% credit against the next monthly support charge.
- **25–100% over target:** 10% credit.
- **>100% over target:** 25% credit AND a written corrective-action plan submitted by TechNexus within 5 business days.

Credits cap at 50% of the monthly support charge.

Channels for raising incidents

Channel	Use for	Hours
`+265 9XX XXX XXX` (on-call mobile)	P1 only	24 / 7
`support@technexusmw.com`	P2, P3, P4	Acknowledged within SLA
WhatsApp Business `+265 9XX XXX XXX`	All severities, citizen support	06:00–22:00
In-portal Help → Report an issue button (planned)	All severities	24 / 7

Every incident receives a ticket number on acknowledgement. Status is visible to MPS through a shared Trello / Linear board (TBD at PO sign).

Exclusions

These SLAs do **not** cover:

- Outages caused by Malawi national power-grid disruption beyond the 30-minute UPS window (unless TechNexus has been contracted for solar back-up under Section E of the quotation).
- Outages caused by telco partners (Airtel, TNM) outside TechNexus's control. TechNexus will assist with diagnosis and escalation to the affected operator.
- Outages during scheduled maintenance windows announced 48 h in advance.
- Force-majeure events.

Quarterly SLA review

A formal SLA-performance report is delivered to MPS the first business day of each quarter. It contains:

1. Incident count by severity
2. Mean response, mean resolution
3. % within target
4. Credit accrual (if any)
5. Trend vs previous quarter

6. Top 3 recurring failure modes + remediation plan

The report is the basis for the quarterly steering committee.



Quarterly Tabletop Exercises

Quarterly Tabletop Exercises

Cadence: Q1, Q2, Q3, Q4 — one scenario per quarter. **Format:** 90 minutes. Facilitator (rotated), scribe (fixed), participants (TechNexus on-call rotation + MPS ICT liaison + Pilot Champion). **Output:** a one-page lessons-learned note filed under docs/operations/lessons/{date}-{scenario}.md, plus any corrective actions logged as engineering issues.

These four scenarios cover the failure modes most likely to arrive at NexusFine first. Run them in rotation; introduce new scenarios as live incidents teach us new ones.

Scenario 1 — HQ outage during business hours

Setting

Friday 14:30 CAT. The Lilongwe Azure region is unreachable for the HQ API. Five officer devices are on shift across Area 18 station. A citizen has just attempted to pay a fine via WhatsApp.

Inject sequence

- T+0 · The on-call engineer's pager fires: "HQ heartbeat lost, last seen 2 min ago."
- T+2 · Pilot Champion at Area 18 calls in: "WhatsApp says payment failed; the supervisor portal is blank."
- T+5 · A second station (Mzuzu, in roll-out phase) is also affected.
- T+12 · Treasury asks where the daily reconciliation file is.

What we expect to see

- Station servers continue accepting counter cash and queueing operational records. Officer devices continue issuing fines into their local queue.
- The on-call engineer triggers Azure status check + opens an Azure support ticket within 5 minutes.
- The Account Lead sends an "issue acknowledged" message to MPS ICT within 10 minutes.
- Status page on `status.nexusfine.mw` updates within 15 minutes.
- Once HQ comes back, the station outbound queues drain in order; receipts appear in the admin Payments page with the correct timestamps.

Questions for the team

1. Who picks up the on-call pager if the primary engineer is at lunch and not answering?
2. Do we have the Azure support phone number printed?
3. Who decides whether to invoke the Plan-B paper backup?
4. How do we confirm the queue drain completed correctly (no lost records)?
5. What goes in the SLA monthly report?

Scenario 2 — Station Server failure (single site)

Setting

Tuesday 09:15. Power has been out at Mchinji Border for 16 hours; the NUC came back on this morning but the SQLite file is reporting "database is locked" and the office Wi-Fi is up but unstable.

Inject sequence

- T+0 · A patrol officer fails to start their shift — the tablet says "station unreachable." The officer phones the station commander.
- T+10 · The station commander tries the supervisor admin URL — connection refused.
- T+25 · Three more officers report the same. A bus has just been pulled over for overloading; the officer has to write a paper ticket.

What we expect to see

- Tablets fall back to paper-ticket mode (officers physically hold a paper book; serial range was logged at deploy time, see site-survey checklist).
- Field engineer dispatched within 90 minutes; spare NUC available at Lilongwe HQ ready to ship.
- Recovery procedure executed: pull the SQLite file from station to HQ, inspect, repair OR replace. No data loss expected because every record was already replicated to HQ on the morning sync.

Questions for the team

1. Does the station commander know the on-call procedure WITHOUT looking it up?
 2. How long do paper tickets need to be retained before being archived into NexusFine after recovery?
 3. Do we have the SQLite recovery commands documented and tested?
 4. Is the spare NUC pre-imaged or does it need a fresh deploy?
-

Scenario 3 — Officer device theft / loss

Setting

Sunday 18:00. An officer reports their device stolen at a roadblock outside Mzuzu CBD. The device was paired to STN-MZU-001 yesterday. The officer had shift-started but had not yet issued any fines today.

Inject sequence

- T+0 · The officer calls the station commander.
- T+30 · The commander dials the on-call engineer via WhatsApp.
- T+2h · The engineer needs to revoke the device certificate AND inform MPS ICT for the police case.

What we expect to see

- The on-call engineer revokes the device via `POST /api/devices/{serial}/revoke` (or its admin-UI equivalent).`
- The audit log records the revocation event with who/when/IP.
- A police case is opened by the station commander — TechNexus assists by providing the device's last-known GPS from the heartbeat history.
- The device is automatically blocked from posting to its station server (mTLS cert revoked).
- 72-hour auto-revoke would also have caught it on Wednesday; manual revocation accelerates the timeline.

Questions for the team

1. Who is authorised to revoke a device?
 2. Is the revocation endpoint protected against social-engineering?
 3. What evidence do we hand to the police?
 4. Replacement device — how fast can the station get a paired spare?
 5. Does revocation cascade to in-flight queued records?
-

Scenario 4 — Suspected supervisor-credential compromise

Setting

Wednesday 11:00. The CISO function notices an anomalous pattern: the supervisor account `super.li` logged in from an IP address in Lagos 90 minutes ago. The supervisor in question is currently in a station meeting in Lilongwe.

Inject sequence

- T+0 · CISO alert: "anomalous login geography."
- T+5 · The supervisor confirms they did not log in from anywhere this morning.
- T+10 · The on-call engineer must decide whether to force a session lock, rotate credentials, or both.

What we expect to see

- The compromised account is immediately disabled.
- All active sessions for that account are invalidated (revoke the JWT signing key, OR force a token-version bump on the user's row).
- Audit log queried for everything that account has done in the last 24 hours; any unauthorised changes flagged and rolled back.
- New temporary credentials issued through HQ IT helpdesk (per the documented credential-reset flow).
- 2FA enforcement (S9, planned) would have stopped this entirely.

Questions for the team

1. Do we have a "kill the JWT signing key" emergency procedure?
 2. How quickly can we read the audit log for one user across all entities?
 3. What do we tell the MPS Cybersecurity Unit, and when?
 4. Is there a public statement required (Data Protection Act §28)?
 5. How do we know it wasn't just a VPN exit node? Confidence threshold?
-

Running the exercise

1. ****Pick a date**** at least a week in advance. Block 90 minutes on calendars.
2. ****Pick a facilitator**** who hasn't run the last one.
3. ****Send the scenario**** to participants the morning of the exercise. Don't send it earlier — we want to test reactions, not preparation.
4. ****Run it live****. The facilitator reads injects out loud at the times listed. Participants act / answer in real time. The scribe captures actions and gaps.
5. ****Debrief immediately**** — 15 minutes at the end. What worked, what didn't, what's missing.
6. ****Write up**** — one-page lessons learned under ``docs/operations/lessons/``.

Open issues for any corrective actions.

A scenario that runs perfectly to plan is suspicious. We expect at least two gaps per exercise. Closing them is the value of doing it.

9 Design — Append-Only Audit Log

Design — Append-Only Audit Log

Status: Design v1.0 (not yet implemented in code) **Owner:** Engineering · reviewed by CISO function **Target wave:** Wave 2 — Module 7 hardening **Decision basis:** D-009 (audit log strategy), S7 in docs/department-recommendations.md

This document records the design TechNexus will implement to make the NexusFine audit log defensible in court and in front of the Auditor General. The current `AuditLog` table is operationally adequate; the design below raises it to evidence-grade.

1. Problem statement

Today the audit log is a regular EF Core entity in the same `AppDbContext` as the operational tables. That has three weaknesses for forensic purposes:

1. ****Same DB → same blast radius.**** A SQL-injection or admin-compromise that can change an operational row could also tamper with the audit row that should prove the change.
2. ****Mutable schema.**** EF migrations could in principle update existing audit rows. No tamper-detection.
3. ****No chain.**** Each row is independent; a deletion or rewrite of an entire audit row would leave no trace.

For the demo this is acceptable. For the pilot and beyond — especially once the Auditor General will be looking at these records — we need an audit log that is **append-only**, **separately hosted**, and **chain-hashed** so any retrospective edit is detectable.

2. Goals & non-goals

Goals

- Each audit record can be cryptographically proven to have been written in the order claimed, with no edits since.
- The log lives in a database account that ****cannot perform UPDATE or DELETE**** on the audit table (DB-role enforcement, not just application code).
- An external auditor with read-only access can verify the chain in their own tooling (the hash schema is open).
- Replay of a payload from the chain reproduces the original event byte-for-byte.

Non-goals

- Public-blockchain anchoring (overkill for pilot; revisit at national scale).
- Realtime streaming to an external SIEM (handled separately as part of the SOC 2 readiness work, S3).
- End-to-end encryption between the application and the audit DB. TLS in transit + at-rest encryption at the storage layer is sufficient.

3. High-level architecture

```
[HQ API] ■■■insert■■■ [Audit Log DB]
■
■ read-only role
▼
[Auditor read-replica / SIEM]
```

- The Audit Log DB is a **separate** SQL Server database (e.g. NexusFineAudit) on its own credentials. The HQ API connects to it with a dedicated low-privileged login.
- The application uses a **single stored procedure** to append. That sproc is the only object the writing role can execute. The role has no TABLE- level grants, no DDL, no SELECT on prior rows.
- Rows in `AuditEntry` carry a **link hash** to the previous row, so the log forms a chain. A retrospective edit breaks the chain from that row forward.

4. Schema

```
CREATE TABLE AuditEntry
(
  Id BIGINT IDENTITY PRIMARY KEY,
  Timestamp DATETIMEOFFSET NOT NULL,
  Sequence BIGINT NOT NULL, -- monotonic
  EntityType VARCHAR(40) NOT NULL,
  EntityId BIGINT NULL,
  Action VARCHAR(40) NOT NULL,
  UserId VARCHAR(100) NULL,
  IPAddress VARCHAR(45) NULL,
  UserAgent VARCHAR(255) NULL,
  Payload NVARCHAR(MAX) NULL, -- JSON
  PrevHash BINARY(32) NULL, -- SHA-256
  RowHash BINARY(32) NOT NULL, -- SHA-256
  StationCode VARCHAR(20) NULL,
  SourceTier VARCHAR(8) NOT NULL, -- HQ / Station / Device
  SchemaVersion INT NOT NULL DEFAULT 1
);

CREATE UNIQUE INDEX IX_AuditEntry_Sequence ON AuditEntry(Sequence);
CREATE INDEX IX_AuditEntry_EntityType_EntityId ON AuditEntry(EntityType, EntityId);
CREATE INDEX IX_AuditEntry_Timestamp ON AuditEntry(Timestamp);
```

```
RowHash = SHA256( Sequence || Timestamp.ToUnixMilliseconds() || EntityType || EntityId
|| Action || UserId || Payload || PrevHash )
```

The first row's PrevHash is 32 zero bytes.

Stored procedure (the only write path)

```
CREATE PROCEDURE sp_AuditAppend
@EntityType VARCHAR(40),
@EntityId BIGINT = NULL,
@Action VARCHAR(40),
@UserId VARCHAR(100) = NULL,
@IPAddress VARCHAR(45) = NULL,
@UserAgent VARCHAR(255) = NULL,
@Payload NVARCHAR(MAX) = NULL,
@StationCode VARCHAR(20) = NULL,
@SourceTier VARCHAR(8)
AS
BEGIN
SET NOCOUNT ON;
```

```

DECLARE @Seq BIGINT, @PrevHash BINARY(32), @Ts DATETIMEOFFSET = SYSDATETIMEOFFSET();

SELECT TOP 1 @Seq = Sequence + 1, @PrevHash = RowHash
FROM AuditEntry ORDER BY Sequence DESC;

IF (@Seq IS NULL) BEGIN SET @Seq = 1; SET @PrevHash = 0x0000...; END

-- Compute row hash on-server so the application cannot precompute & forge
DECLARE @Hash BINARY(32) = HASHBYTES('SHA2_256',
CONCAT(@Seq, '|', DATEDIFF_BIG(MS,'1970-01-01',@Ts),'|',
@EntityType,'|',ISNULL(@EntityId,0),'|',@Action,'|',
ISNULL(@UserId,''),'|',ISNULL(@Payload,''),'|',
CAST(@PrevHash AS VARCHAR(64))));

INSERT AuditEntry (Timestamp, Sequence, EntityType, EntityId, Action,
UserId, IPAddress, UserAgent, Payload,
PrevHash, RowHash, StationCode, SourceTier)
VALUES (@Ts, @Seq, @EntityType, @EntityId, @Action,
@UserId, @IpAddress, @UserAgent, @Payload,
@PrevHash, @Hash, @StationCode, @SourceTier);
END

```

Role grants

```

CREATE ROLE audit_writer;
GRANT EXECUTE ON sp_AuditAppend TO audit_writer;
-- NO grants on TABLE AuditEntry to this role.

CREATE ROLE audit_reader;
GRANT SELECT ON AuditEntry TO audit_reader;

```

The HQ API's audit-write SQL login is in audit_writer only. The auditor's SQL login (or read-replica) is in audit_reader only.

5. Verification routine

Auditors verify by walking the chain:

```

DECLARE @prev BINARY(32) = 0x0000...; -- 32 zero bytes
DECLARE @id BIGINT = 0, @bad BIGINT = NULL;
WHILE 1=1
BEGIN
SELECT TOP 1 Id, Sequence, PrevHash, RowHash, ...
FROM AuditEntry WHERE Id > @id ORDER BY Id;
IF @@ROWCOUNT = 0 BREAK;

IF PrevHash <> @prev BEGIN SET @bad = Id; BREAK; END
-- recompute expected RowHash, compare; flag mismatch.

SET @prev = RowHash;
END
SELECT @bad AS firstBrokenRowId;

```

The same routine ships as a CLI tool (tools/audit-verify) for off-server verification.

6. Migration path

Phase 1 — Add the new DB alongside the existing one

- Provision `NexusFineAudit` database.

- Run schema + sparc + roles.
- Add a second connection string `AuditDb` to API config.
- Add an `IAuditAppender` interface in `NexusFine.Infrastructure`.
- Two implementations: `LegacyAuditAppender` (writes to current `AppDbContext.AuditLogs`) and `ChainedAuditAppender` (calls `sp_AuditAppend`).
- Toggle by config flag `Audit:UseChainedLog`.

Phase 2 — Run both in parallel for a fortnight

- `AuditLogMiddleware` calls both appenders.
- Compare row counts daily.

Phase 3 — Cut over

- Switch the feature flag.
- Keep the legacy table read-only for 90 days; then archive.

Phase 4 — Auditor on-boarding

- Provision Auditor General's office with `audit\_reader` credentials on a read-replica.
- Train them on the verification routine.

7. Operational considerations

Consideration	Treatment
Performance	`sp_AuditAppend` is single-row, single-index; benchmarked at >2,000 ops/sec on Azure SQL S2. NexusFine load is <50 ops/sec at national rollout.
Failure mode	If the audit DB is unreachable, the API must fail the originating mutation (no silent skips). A circuit-breaker pattern controls cascading failure.
Backfill / late writes	Late writes (from a station that comes back online after days offline) get a current timestamp + the original "occurred-at" stored in payload.
Schema evolution	`SchemaVersion` column on the row; verifier handles version bumps.
Retention	Audit is never purged at HQ. The 90-day station window applies only to local cached audit; HQ keeps forever.

8. Open questions for the Auditor General

These should be discussed before Phase 1 starts:

1. Is SHA-256 acceptable, or does the office prefer a NIST FIPS 180-4 set?
2. Are there specific events the office wants tagged for accelerated review?
3. Does the office want the verifier tool to be a CLI, a web UI, or a SQL script?
4. What is the office's preferred export format for archive (SQL backup, CSV, JSON, signed bundle)?
5. Does the office want sample monthly reports auto-generated, or only on demand?

9. References

- `docs/DECISIONS.md` — D-009 (audit middleware strategy)
- `docs/security/threat-model.md` — STRIDE row for audit-log tampering
- `docs/compliance/dpia-malawi.md` — §5.7 (audit-log immutability)
- ISO 27001:2022 control 8.15 (Logging)

1 Threat Model (STRIDE)

NexusFine — Threat Model (STRIDE)

Status: Draft v1 · **Owner:** TechNexus CISO function · **Reviewed:** 2026-05-16 **Next review:** quarterly, and after any architecture change **Scope:** Malawi pilot. Same model applies to ZM/MZ/TZ tenants by substitution.

This document classifies threats against NexusFine using STRIDE (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege) and records the controls in place, the residual risk, and the planned hardening path.

1. System decomposition

NexusFine has six trust boundaries:



Data classes:

- ****PII**** — driver name, national ID, licence number, phone, plate
- ****Financial**** — payment amounts, receipts, channel references
- ****Operational**** — fines issued, status, photos, GPS, NFC tag IDs
- ****Audit**** — every state change, write-only, immutable

2. STRIDE analysis (top assets)

2.1 Citizen-portal fine lookup

STRIDE	Threat	Control	Residual
S	Citizen impersonates another driver to view their fines	Public lookup intentionally returns only reference, plate, amount, status — no PII. Detail page gated by a one-time code sent to phone on record.	Low
T	Attacker tampers with lookup response in-flight	TLS 1.2+ enforced on public endpoint. HSTS preload.	Low
R	Disputed lookups	Lookup events captured in audit log with IP + user agent.	Low

STRIDE	Threat	Control	Residual
I	Mass scrape of plates / refs	Rate limiter `public` policy: 120 req/min/IP. WAF rules at deploy.	Medium
D	DDoS against the public site	Cloud WAF (Azure Front Door) + auto-scaling App Service.	Medium
E	Lookup endpoint escalates to admin	`[AllowAnonymous]` only on `lookup`; all admin endpoints require `[Authorize(Roles=...)]`.	Low

2.2 Officer fine issuance (tablet → Station → HQ)

STRIDE	Threat	Control	Residual
S	Stolen device impersonates an officer	Shared-device login switch; 72-hour heartbeat → auto-revoke. Device cert pinned at station.	Medium
T	Tampering with offline-queued fines	Each record carries a client-UUID idempotency key; HQ rejects on mismatch with cryptographic hash on push.	Medium
R	Officer denies issuing a fine	Every issuance records badge, GPS, timestamp, photo, NFC scan — all in audit log.	Low
I	Citizen PII captured on a tablet exposed to others	Tablet PIN; screen auto-locks after 60s; SQLCipher on the local cache (planned S5).	Medium
D	Tablet flood pushes fake fines to station	Per-device throttle at station (planned).	Medium
E	Officer obtains supervisor token	JWT roles enforced server-side; supervisor 2FA (planned S9).	Low

2.3 Payment posting

STRIDE	Threat	Control	Residual
S	Forged gateway callback marks a fine as paid	Callback verified against gateway's HMAC signature. Simulator clearly labelled for demo.	Low

STRIDE	Threat	Control	Residual
T	In-flight tampering with payment amount	TLS to gateways. Server-side recompute of amount from fine.id + offence tariff at confirm time.	Low
R	Citizen denies authorising payment	All initiates logged with phone + IP + channel; gateway TX reference stored verbatim.	Low
I	Receipt leaks PII	Receipt template carries only plate, reference, amount, channel — no national ID or address.	Low
D	Spurious "pay" floods	Rate limiter on <code>/api/payments/initiate`</code> (treat as <code>`public`</code>); idempotency via fine-ref + channel + window.	Low
E	Payment endpoint accepts unauth state change	Initiate is intentionally <code>`[AllowAnonymous]`</code> (citizen path) but never sets <code>`Fine.Status`</code> — only <code>`Confirm`</code> flow does that, and the simulator mirrors the prod HMAC check.	Low

2.4 Admin portal (supervisor / admin)

STRIDE	Threat	Control	Residual
S	Credential stuffing on <code>/api/auth/login`</code>	Rate limiter <code>`sensitive`</code> : 10 req/min/IP. 2FA at supervisor + admin (planned S9). Account-lock-out after 10 failures (planned).	Medium
T	XSS / CSRF	Blazor Server is server-rendered; <code>`app.UseAntiforgery()`</code> set. No <code>`innerHTML`</code> usage.	Low
R	Admin denies running a destructive action	All POST/PUT/PATCH/DELETE captured by <code>`AuditLogMiddleware`</code> with userId, IP, before/after JSON.	Low
I	JWT theft	JWT held only in Blazor circuit memory (D-011). Page reload kicks user back to <code>/login`</code> .	Medium

STRIDE	Threat	Control	Residual
D	Login flood	`sensitive` rate-limit window.	Low
E	Supervisor accesses Admin-only CRUD	All write endpoints carry `[A authorize(Roles="Admin")]`; UI hides write buttons via ` <authorizeview >`.<="" roles="Admin" td=""><td>Low</td></authorizeview>	Low

2.5 Station Server ↔ HQ sync

STRIDE	Threat	Control	Residual
S	Rogue station server posts fake fines to HQ	mTLS with per-station cert pinned in `Station.StationServerPublicKey` (planned S2).	High in current build
T	Replay of operational push	ClientUuid idempotency key on every record; HQ dedups.	Low
R	Station disputes records it sent	Sync events recorded both ends with payload digest.	Low
I	Reference data leakage on station theft	SQLCipher on the station DB (planned S5). Disk-level encryption (BitLocker / LUKS) on NUC.	Medium
D	HQ unreachable	Designed-for graceful degradation: stations operate fully offline; outbound queue retries up to 7 days.	Low
E	Station server escalates to HQ admin	Station's cert grants only the `station` role at HQ; no admin scope.	Low

3. Top residual risks (sorted by demo-day exposure)

#	Risk	Owner	Target close date
1	mTLS not yet enforced HQ ↔ Station	CISO	Before pilot go-live
2	Station SQLite is plain-text	CISO	Wave 2 (S5)
3	No 2FA at supervisor/admin level	CISO	Wave 4 (S9)
4	No WAF in front of public endpoints	CISO + Ops	Pilot deploy phase
5	NoOpAuthHandler is permissive (admin HTTP layer)	Engineering	Replace once we have a real cookie + session strategy

4. Controls inventory (what's already in place)

- ASP.NET Core JWT auth (HMAC-SHA256, configurable secret, 120-min TTL)
- `AuditLogMiddleware` records every mutation
- `JsonStringEnumConverter` globally — prevents enum-as-int payload tampering
- Rate-limiter middleware (`public` / `sensitive` policies)
- Collision-safe receipt and reference generation (no `Count() + 1` race window)
- HTTPS redirection middleware
- Antiforgery middleware (Blazor)
- Role-based authorisation on every write endpoint
- One-time, hashed pairing tokens for tablet enrolment
- Heartbeat-based device revocation (72-h policy)

5. Change log

Date	Change
2026-05-16	Initial draft

1 OSS Licence Report (CSV)

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Admin.csproj	Azure.Core	1.47.1			UNKNOWN
NexusFine.Admin.csproj	Azure.Identity	1.14.2			UNKNOWN
NexusFine.Admin.csproj	Microsoft.AspNetCore.Authorization	10.0.5			UNKNOWN
NexusFine.Admin.csproj	Microsoft.Bcl.AsyncInterfaces	8.0.0			UNKNOWN
NexusFine.Admin.csproj	Microsoft.Bcl.Cryptography	9.0.4			UNKNOWN
NexusFine.Admin.csproj	Microsoft.Data.SqlClient	6.1.1			UNKNOWN
NexusFine.Admin.csproj	Microsoft.Data.SqlClient.SNI.runtime	6.0.2			UNKNOWN
NexusFine.Admin.csproj	Microsoft.EntityFrameworkCore	10.0.5			UNKNOWN
NexusFine.Admin.csproj	Microsoft.EntityFrameworkCore.Abstractions	10.0.5			UNKNOWN
NexusFine.Admin.csproj	Microsoft.EntityFrameworkCore.Analyzers	10.0.5			UNKNOWN
NexusFine.Admin.csproj	Microsoft.EntityFrameworkCore.Relational	10.0.5			UNKNOWN
NexusFine.Admin.csproj	Microsoft.EntityFrameworkCore.SqlServer	10.0.5			UNKNOWN
NexusFine.Admin.csproj	Microsoft.Identity.Client	4.73.1			UNKNOWN
NexusFine.Admin.csproj	Microsoft.Identity.Client.Extensions.Msal	4.73.1			UNKNOWN
NexusFine.Admin.csproj	Microsoft.IdentityModel.Abstractions	8.2.1			UNKNOWN
NexusFine.Admin.csproj	Microsoft.IdentityModel.JsonWebTokens	8.2.1			UNKNOWN

Project	Package	Version	License	ProjectUrl	Status
NexusFine.Admin.csproj	Microsoft.IdentityModel.Logging	8.2.1			UNKNOWN
NexusFine.Admin.csproj	Microsoft.IdentityModel.Protocols	7.7.1			UNKNOWN
NexusFine.Admin.csproj	Microsoft.IdentityModel.Protocols.OpenIdConnect	7.7.1			UNKNOWN
NexusFine.Admin.csproj	Microsoft.IdentityModel.Tokens	8.2.1			UNKNOWN
NexusFine.Admin.csproj	Microsoft.SqlServer.Server	1.0.0			UNKNOWN
NexusFine.Admin.csproj	System.ClientModel	1.5.1			UNKNOWN
NexusFine.Admin.csproj	System.Configuration.ConfigurationManager	9.0.4			UNKNOWN
NexusFine.Admin.csproj	System.IdentityModel.Tokens.Jwt	8.2.1			UNKNOWN
NexusFine.Admin.csproj	System.Memory.Data	8.0.1			UNKNOWN
NexusFine.Admin.csproj	System.Security.Cryptography.Pkcs	9.0.4			UNKNOWN
NexusFine.Admin.csproj	System.Security.Cryptography.ProtectedData	9.0.4			UNKNOWN
NexusFine.API.csproj	Azure.Core	1.47.1			UNKNOWN
NexusFine.API.csproj	Azure.Identity	1.14.2			UNKNOWN
NexusFine.API.csproj	Humanizer.Core	2.14.1			UNKNOWN
NexusFine.API.csproj	Microsoft.AspNetCore.Authentication.JwtBearer	10.0.5			UNKNOWN
NexusFine.API.csproj	Microsoft.Bcl.AsyncInterfaces	8.0.0			UNKNOWN
NexusFine.API.csproj	Microsoft.Bcl.Cryptography	9.0.4			UNKNOWN
NexusFine.API.csproj	Microsoft.Build.Framework	18.0.2			UNKNOWN
NexusFine.API.csproj	Microsoft.CodeAnalysis.Analyzers	3.11.0			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.API.c sproj	Microsoft.CodeA nalysis.Common	5.0.0			UNKNOWN
NexusFine.API.c sproj	Microsoft.CodeA nalysis.CSharp	5.0.0			UNKNOWN
NexusFine.API.c sproj	Microsoft.CodeA nalysis.CSharp. Workspaces	5.0.0			UNKNOWN
NexusFine.API.c sproj	Microsoft.CodeA nalysis.Workspac es.Common	5.0.0			UNKNOWN
NexusFine.API.c sproj	Microsoft.CodeA nalysis.Workspac es.MSBuild	5.0.0			UNKNOWN
NexusFine.API.c sproj	Microsoft.Data.S qlClient	6.1.1			UNKNOWN
NexusFine.API.c sproj	Microsoft.Data.S qlClient.SNI.runti me	6.0.2			UNKNOWN
NexusFine.API.c sproj	Microsoft.EntityFr ameworkCore	10.0.5			UNKNOWN
NexusFine.API.c sproj	Microsoft.EntityFr ameworkCore.Ab stractions	10.0.5			UNKNOWN
NexusFine.API.c sproj	Microsoft.EntityFr ameworkCore.An alyzers	10.0.5			UNKNOWN
NexusFine.API.c sproj	Microsoft.EntityFr ameworkCore.De sign	10.0.5			UNKNOWN
NexusFine.API.c sproj	Microsoft.EntityFr ameworkCore.Re lational	10.0.5			UNKNOWN
NexusFine.API.c sproj	Microsoft.EntityFr ameworkCore.Sq lServer	10.0.5			UNKNOWN
NexusFine.API.c sproj	Microsoft.Extensi ons.ApiDescriptio n.Server	6.0.5			UNKNOWN
NexusFine.API.c sproj	Microsoft.Extensi ons.Dependency Model	10.0.5			UNKNOWN
NexusFine.API.c sproj	Microsoft.Identity .Client	4.73.1			UNKNOWN
NexusFine.API.c sproj	Microsoft.Identity .Client.Extension s.Msal	4.73.1			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.API.csproj	Microsoft.IdentityModel.Abstractions	8.2.1			UNKNOWN
NexusFine.API.csproj	Microsoft.IdentityModel.JsonWebTokens	8.2.1			UNKNOWN
NexusFine.API.csproj	Microsoft.IdentityModel.Logging	8.2.1			UNKNOWN
NexusFine.API.csproj	Microsoft.IdentityModel.Protocols	8.0.1			UNKNOWN
NexusFine.API.csproj	Microsoft.IdentityModel.Protocols.OpenIdConnect	8.0.1			UNKNOWN
NexusFine.API.csproj	Microsoft.IdentityModel.Tokens	8.2.1			UNKNOWN
NexusFine.API.csproj	Microsoft.OpenApi	1.6.22			UNKNOWN
NexusFine.API.csproj	Microsoft.SqlServer.Server	1.0.0			UNKNOWN
NexusFine.API.csproj	Microsoft.VisualStudio.SolutionPersistence	1.0.52			UNKNOWN
NexusFine.API.csproj	Mono.TextTemplating	3.0.0			UNKNOWN
NexusFine.API.csproj	Newtonsoft.Json	13.0.3			UNKNOWN
NexusFine.API.csproj	Swashbuckle.AspNetCore	6.9.0			UNKNOWN
NexusFine.API.csproj	Swashbuckle.AspNetCore.Swagger	6.9.0			UNKNOWN
NexusFine.API.csproj	Swashbuckle.AspNetCore.SwaggerGen	6.9.0			UNKNOWN
NexusFine.API.csproj	Swashbuckle.AspNetCore.SwaggerUI	6.9.0			UNKNOWN
NexusFine.API.csproj	System.ClientModel	1.5.1			UNKNOWN
NexusFine.API.csproj	System.CodeDom	6.0.0			UNKNOWN
NexusFine.API.csproj	System.Composition	9.0.0			UNKNOWN

Project	Package	Version	License	ProjectUrl	Status
NexusFine.API.csproj	System.Composition.AttributedModel	9.0.0			UNKNOWN
NexusFine.API.csproj	System.Composition.Convention	9.0.0			UNKNOWN
NexusFine.API.csproj	System.Composition.Hosting	9.0.0			UNKNOWN
NexusFine.API.csproj	System.Composition.Runtime	9.0.0			UNKNOWN
NexusFine.API.csproj	System.Composition.TypedParts	9.0.0			UNKNOWN
NexusFine.API.csproj	System.Configuration.ConfigurationManager	9.0.4			UNKNOWN
NexusFine.API.csproj	System.IdentityModel.Tokens.Jwt	8.2.1			UNKNOWN
NexusFine.API.csproj	System.Memory.Data	8.0.1			UNKNOWN
NexusFine.API.csproj	System.Security.Cryptography.Pkcs	9.0.4			UNKNOWN
NexusFine.API.csproj	System.Security.Cryptography.ProtectedData	9.0.4			UNKNOWN
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NexusFine.Infras tructure.csproj	Azure.Identity	1.14.2			UNKNOWN
NexusFine.Infras tructure.csproj	Humanizer.Core	2.14.1			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Bcl.AsyncInterfaces	8.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Bcl.Cryptography	9.0.4			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Build.Fram ework	18.0.2			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.CodeAnalysis.Analyzers	3.11.0			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.CodeAnalysis.Common	5.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.CodeAnalysis.CSharp	5.0.0			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
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NexusFine.Infras tructure.csproj	Microsoft.CodeA nalysis.Workspac es.MSBuild	5.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Data.S qlClient	6.1.1			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Data.S qlClient.SNI.runti me	6.0.2			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.EntityFr ameworkCore	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.EntityFr ameworkCore.Ab stractions	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.EntityFr ameworkCore.An alyzers	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.EntityFr ameworkCore.De sign	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.EntityFr ameworkCore.Re lational	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.EntityFr ameworkCore.Sq lServer	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.EntityFr ameworkCore.To ols	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Caching.Abs tractions	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Caching.Me mory	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Configuratio n	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Configuratio n.Abstractions	10.0.5			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Configuratio n.Binder	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.DependencyI njection	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.DependencyI njection.Abstracti ons	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Dependency Model	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Diagnostics	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Diagnostics. Abstractions	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Http	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Logging	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Logging.Abst ractions	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Options	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Options.Conf igurationExtensio ns	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Extensi ons.Primitives	10.0.5			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Identity .Client	4.73.1			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Identity .Client.Extension s.Msal	4.73.1			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Identity Model.Abtractio ns	8.2.1			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Identity Model.JsonWebT okens	8.2.1			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Identity Model.Logging	8.2.1			UNKNOWN

Project	Package	Version	License	ProjectUrl	Status
NexusFine.Infras tructure.csproj	Microsoft.Identity Model.Protocols	7.7.1			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Identity Model.Protocols. OpenIdConnect	7.7.1			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.Identity Model.Tokens	8.2.1			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.SqlSer ver.Server	1.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	Microsoft.VisualStudio.SolutionPer sistence	1.0.52			UNKNOWN
NexusFine.Infras tructure.csproj	Mono.TextTempl ating	3.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	Newtonsoft.Json	13.0.3			UNKNOWN
NexusFine.Infras tructure.csproj	System.ClientMo del	1.5.1			UNKNOWN
NexusFine.Infras tructure.csproj	System.CodeDo m	6.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	System.Composi tion	9.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	System.Composi tion.AttributedMo del	9.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	System.Composi tion.Convention	9.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	System.Composi tion.Hosting	9.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	System.Composi tion.Runtime	9.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	System.Composi tion.TypedParts	9.0.0			UNKNOWN
NexusFine.Infras tructure.csproj	System.Configur ation.Configurati onManager	9.0.4			UNKNOWN
NexusFine.Infras tructure.csproj	System.Diagnosti cs.EventLog	9.0.4			UNKNOWN
NexusFine.Infras tructure.csproj	System.IdentityM odel.Tokens.Jwt	8.2.1			UNKNOWN
NexusFine.Infras tructure.csproj	System.Memory. Data	8.0.1			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Infras tructure.csproj	System.Security. Cryptography.Pk cs	9.0.4			UNKNOWN
NexusFine.Infras tructure.csproj	System.Security. Cryptography.Pr otectedData	9.0.4			UNKNOWN
NexusFine.Mobil e.csproj	GoogleGson	2.13.1.1			UNKNOWN
NexusFine.Mobil e.csproj	Microsoft.Extensi ons.Configuratio n	10.0.5			UNKNOWN
NexusFine.Mobil e.csproj	Microsoft.Extensi ons.Configuratio n	10.0.5			UNKNOWN
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NexusFine.Mobil e.csproj	Microsoft.Extensi ons.Configuratio n.Abstractions	10.0.5			UNKNOWN
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NexusFine.Mobil e.csproj	Microsoft.Extensi ons.Configuratio n.Binder	10.0.5			UNKNOWN
NexusFine.Mobil e.csproj	Microsoft.Extensi ons.Configuratio n.Binder	10.0.5			UNKNOWN
NexusFine.Mobil e.csproj	Microsoft.Extensi ons.Configuratio n.Binder	10.0.5			UNKNOWN
NexusFine.Mobil e.csproj	Microsoft.Extensi ons.Configuratio n.Binder	10.0.5			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Microsoft.Extensions.DependencyInjection	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.DependencyInjection	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.DependencyInjection	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.DependencyInjection	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.DependencyInjection.Abstractions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.DependencyInjection.Abstractions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.DependencyInjection.Abstractions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.DependencyInjection.Abstractions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Diagnostics	10.0.5			UNKNOWN
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NexusFine.Mobile.csproj	Microsoft.Extensions.Diagnostics	10.0.5			UNKNOWN
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NexusFine.Mobile.csproj	Microsoft.Extensions.Diagnostics.Abstractions	10.0.5			UNKNOWN
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NexusFine.Mobile.csproj	Microsoft.Extensions.Diagnostics.Abstractions	10.0.5			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
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NexusFine.Mobile.csproj	Microsoft.Extensions.FileProviders.Abstractions	10.0.0			UNKNOWN
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NexusFine.Mobile.csproj	Microsoft.Extensions.Hosting.Abstractions	10.0.0			UNKNOWN
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NexusFine.Mobile.csproj	Microsoft.Extensions.Http	10.0.5			UNKNOWN
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NexusFine.Mobile.csproj	Microsoft.Extensions.Logging	10.0.5			UNKNOWN
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NexusFine.Mobile.csproj	Microsoft.Extensions.Logging	10.0.5			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Microsoft.Extensions.Logging.Abstractions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Logging.Abstractions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Logging.Abstractions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Logging.Abstractions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Logging.Debug	10.0.0			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Logging.Debug	10.0.0			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Logging.Debug	10.0.0			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Logging.Debug	10.0.0			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Options	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Options	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Options	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Options	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Options.ConfigurationExtensions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Options.ConfigurationExtensions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Options.ConfigurationExtensions	10.0.5			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Microsoft.Extensions.Options.ConfigurationExtensions	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Primitives	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Primitives	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Primitives	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Extensions.Primitives	10.0.5			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Graphics.Win2D	1.3.2			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.IO.RecyclableMemoryStream	3.0.1			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Build.Tasks	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Build.Tasks	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Build.Tasks	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Build.Tasks	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Core	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Core	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Core	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Core	10.0.20			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Xaml	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Xaml	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Xaml	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Controls.Xaml	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Core	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Core	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Core	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Core	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Essentials	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Essentials	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Essentials	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Essentials	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Graphics	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Graphics	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Graphics	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Graphics	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Graphics.Win2D.WinUI.Desktop	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Resizetizer	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Resizetizer	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Resizetizer	10.0.20			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Maui.Resizetizer	10.0.20			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Microsoft.Web.WebView2	1.0.3179.45			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.Windows.SDK.BuildTools	10.0.22621.756			UNKNOWN
NexusFine.Mobile.csproj	Microsoft.WindowsAppSDK	1.7.250909003			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Android.Glide	4.16.0.14			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Android.Glide.Annotations	4.16.0.14			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Android.Glide.DiskLruCache	4.16.0.14			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Android.Glide.GifDecoder	4.16.0.14			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Activity	1.10.1.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Activity.Ktx	1.10.1.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Annotation	1.9.1.5			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Annotation.Experimental	1.5.1.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Annotation.Jvm	1.9.1.5			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.AppCompat	1.7.1.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.AppCompat.AppCompatResources	1.7.1.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Arch.Core.Common	2.2.0.18			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Arch.Core.Runtime	2.2.0.18			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Browser	1.8.0.11			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.CardView	1.0.0.36			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Xamarin.AndroidX.Collection	1.5.0.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Collection.Jvm	1.5.0.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Collection.Ktx	1.5.0.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Concurrent.Futures	1.3.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.ConstraintLayout	2.2.1.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.ConstraintLayout.Core	1.1.1.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.CoordinatorLayout	1.3.0.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Core	1.16.0.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Core.Core.Ktx	1.16.0.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Core.ViewTree	1.0.0.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.CursorAdapter	1.0.0.34			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.CustomView	1.2.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.CustomView.PoolingContainer	1.1.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.DrawerLayout	1.2.0.18			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.DynamicAnimation	1.1.0.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Emoji2	1.5.0.6			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Emoji2.ViewsHelper	1.5.0.6			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.ExifInterface	1.4.1.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Fragment	1.8.8.1			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Xamarin.AndroidX.Fragment.Ktx	1.8.8.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Interpolator	1.0.0.34			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.Common	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.Common.Jvm	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.LiveData	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.LiveData.Core	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.LiveData.Core.Ktx	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.Process	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.Runtime	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.Runtime.Android	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.Runtime.Ktx	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.Runtime.Ktx.Android	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.ViewModel	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.ViewModel.Android	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.ViewModel.Ktx	2.9.2.1			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.ViewModelSavedState	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Lifecycle.ViewModelSavedState.Android	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Loader	1.1.0.34			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Navigation.Common	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Navigation.Common.Android	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Navigation.Fragment	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Navigation.Runtime	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Navigation.Runtime.Android	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Navigation.UI	2.9.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.ProfileInstaller.ProfileInstaller	1.4.1.5			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.RecyclerView	1.4.0.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.ResourceInspection.Annotation	1.0.1.22			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.SavedState	1.3.1.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.SavedState.SavedState.Android	1.3.1.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.SavedState.SavedState.Ktx	1.3.1.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Security.SecurityCrypto	1.1.0.4-alpha07			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Xamarin.AndroidX.SlidingPaneLayout	1.2.0.22			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Startup.StartupRuntime	1.2.0.5			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.SwipeRefreshLayout	1.1.0.29			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Tracing.Tracing	1.3.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Tracing.Tracing.Android	1.3.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Transition	1.6.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.VectorDrawable	1.2.0.8			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.VectorDrawableAnimated	1.2.0.8			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.VersionedParcelable	1.2.1.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.ViewPager	1.1.0.4			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.ViewPager2	1.1.0.8			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Window	1.4.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Window.WindowCore	1.4.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.AndroidX.Window.WindowCore.Jvm	1.4.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Google.Android.Material	1.12.0.5			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Google.Code.FindBugs.JSR305	3.0.2.21			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Google.Crypto.Tink.Android	1.18.0.1			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Mobile.csproj	Xamarin.Google. ErrorProne.Annotations	2.41.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Google. Guava.Listenable Future	1.0.0.29			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Jetbrains. Annotations	26.0.2.3			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.JSpecify	1.0.0.4			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.Kotlin.St dLib	2.2.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.KotlinX. Coroutines.Andro id	1.10.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.KotlinX. Coroutines.Core	1.10.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.KotlinX. Coroutines.Core. Jvm	1.10.2.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.KotlinX. Serialization.Cor e	1.9.0.1			UNKNOWN
NexusFine.Mobile.csproj	Xamarin.KotlinX. Serialization.Cor e.Jvm	1.9.0.1			UNKNOWN
NexusFine.Station.csproj	Humanizer.Core	2.14.1			UNKNOWN
NexusFine.Station.csproj	Microsoft.Build.Fr amework	18.0.2			UNKNOWN
NexusFine.Station.csproj	Microsoft.CodeA nalysis.Analyzers	3.11.0			UNKNOWN
NexusFine.Station.csproj	Microsoft.CodeA nalysis.Common	5.0.0			UNKNOWN
NexusFine.Station.csproj	Microsoft.CodeA nalysis.CSharp	5.0.0			UNKNOWN
NexusFine.Station.csproj	Microsoft.CodeA nalysis.CSharp. Workspaces	5.0.0			UNKNOWN
NexusFine.Station.csproj	Microsoft.CodeA nalysis.Workspac es.Common	5.0.0			UNKNOWN
NexusFine.Station.csproj	Microsoft.CodeA nalysis.Workspac es.MSBuild	5.0.0			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Station.csproj	Microsoft.Data.Sqlite.Core	10.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.EntityFrameworkCore	10.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.EntityFrameworkCore.Abstractions	10.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.EntityFrameworkCore.Analyzers	10.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.EntityFrameworkCore.Design	10.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.EntityFrameworkCore.Relational	10.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.EntityFrameworkCore.Sqlite	10.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.EntityFrameworkCore.Sqlite.Core	10.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.Extensions.ApiDescription.Server	6.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.Extensions.DependencyModel	10.0.5			UNKNOWN
NexusFine.Station.csproj	Microsoft.OpenApi	1.6.14			UNKNOWN
NexusFine.Station.csproj	Microsoft.VisualStudio.SolutionPersistence	1.0.52			UNKNOWN
NexusFine.Station.csproj	Mono.TextTemplating	3.0.0			UNKNOWN
NexusFine.Station.csproj	Newtonsoft.Json	13.0.3			UNKNOWN
NexusFine.Station.csproj	SQLitePCLRaw.bundle_e_sqlite3	2.1.11			UNKNOWN
NexusFine.Station.csproj	SQLitePCLRaw.core	2.1.11			UNKNOWN
NexusFine.Station.csproj	SQLitePCLRaw.lib.e_sqlite3	2.1.11			UNKNOWN

Project	Package	Version	License	ProjectUrl	Status
NexusFine.Station.csproj	SQLitePCLRaw.provider.e_sqlite3	2.1.11			UNKNOWN
NexusFine.Station.csproj	Swashbuckle.AspNetCore	6.9.0			UNKNOWN
NexusFine.Station.csproj	Swashbuckle.AspNetCore.Swagger	6.9.0			UNKNOWN
NexusFine.Station.csproj	Swashbuckle.AspNetCore.SwaggerGen	6.9.0			UNKNOWN
NexusFine.Station.csproj	Swashbuckle.AspNetCore.SwaggerUI	6.9.0			UNKNOWN
NexusFine.Station.csproj	System.CodeDom	6.0.0			UNKNOWN
NexusFine.Station.csproj	System.Composition	9.0.0			UNKNOWN
NexusFine.Station.csproj	System.Composition.AttributedModel	9.0.0			UNKNOWN
NexusFine.Station.csproj	System.Composition.Convention	9.0.0			UNKNOWN
NexusFine.Station.csproj	System.Composition.Hosting	9.0.0			UNKNOWN
NexusFine.Station.csproj	System.Composition.Runtime	9.0.0			UNKNOWN
NexusFine.Station.csproj	System.Composition.TypedParts	9.0.0			UNKNOWN
NexusFine.Tests.csproj	Azure.Core	1.47.1			UNKNOWN
NexusFine.Tests.csproj	Azure.Identity	1.14.2			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Bcl.AsyncInterfaces	8.0.0			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Bcl.Cryptography	9.0.4			UNKNOWN
NexusFine.Tests.csproj	Microsoft.CodeCoverage	17.11.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Data.SqlClient	6.1.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Data.SqlClient.SNI.runtime	6.0.2			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Tests.csproj	Microsoft.EntityFrameworkCore	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.EntityFrameworkCore.Abstractions	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.EntityFrameworkCore.Analyzers	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.EntityFrameworkCore.InMemory	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.EntityFrameworkCore.Relational	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.EntityFrameworkCore.SqlServer	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Caching.Abstractions	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Caching.Memory	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Configuration	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Configuration.Abstractions	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Configuration.Binder	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.DependencyInjection	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.DependencyInjection.Abstractions	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Diagnostics	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Diagnostics.Abstractions	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Http	10.0.5			UNKNOWN

Project	Package	Version	License	ProjectUrl	Status
NexusFine.Tests.csproj	Microsoft.Extensions.Logging	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Logging.Abstractions	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Options	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Options.ConfigurationExtensions	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Extensions.Primitives	10.0.5			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Identity.Client	4.73.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.Identity.Client.Extensions.Msal	4.73.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.IdentityModel.Abstractions	8.2.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.IdentityModel.JsonWebTokens	8.2.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.IdentityModel.Logging	8.2.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.IdentityModel.Protocols	7.7.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.IdentityModel.Protocols.OpenIdConnect	7.7.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.IdentityModel.Tokens	8.2.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.NET.Test.Sdk	17.11.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.SqlServer.Server	1.0.0			UNKNOWN
NexusFine.Tests.csproj	Microsoft.TestPlatform.ObjectModel	17.11.1			UNKNOWN
NexusFine.Tests.csproj	Microsoft.TestPlatform.TestHost	17.11.1			UNKNOWN
NexusFine.Tests.csproj	Newtonsoft.Json	13.0.1			UNKNOWN

Project	Package	Version	Licence	ProjectUrl	Status
NexusFine.Tests.csproj	System.ClientModel	1.5.1			UNKNOWN
NexusFine.Tests.csproj	System.Configuration.ConfigurationManager	9.0.4			UNKNOWN
NexusFine.Tests.csproj	System.Diagnostics.EventLog	9.0.4			UNKNOWN
NexusFine.Tests.csproj	System.IdentityModel.Tokens.Jwt	8.2.1			UNKNOWN
NexusFine.Tests.csproj	System.Memory.Data	8.0.1			UNKNOWN
NexusFine.Tests.csproj	System.Security.Cryptography.Pkcs	9.0.4			UNKNOWN
NexusFine.Tests.csproj	System.Security.Cryptography.ProtectedData	9.0.4			UNKNOWN
NexusFine.Tests.csproj	xunit	2.9.2			UNKNOWN
NexusFine.Tests.csproj	xunit.abstractions	2.0.3			UNKNOWN
NexusFine.Tests.csproj	xunit.analyzers	1.16.0			UNKNOWN
NexusFine.Tests.csproj	xunit.assert	2.9.2			UNKNOWN
NexusFine.Tests.csproj	xunit.core	2.9.2			UNKNOWN
NexusFine.Tests.csproj	xunit.extensibility.core	2.9.2			UNKNOWN
NexusFine.Tests.csproj	xunit.extensibility.execution	2.9.2			UNKNOWN
NexusFine.Tests.csproj	xunit.runner.visualstudio	2.8.2			UNKNOWN